SRI CHANDRASEKHARENDRA SARASWATHI VISWA MAHAVIDYALAYA

(UNIVERSITY ESTABLISHED UNDER SECTION 3 OF UGC ACT 1956)
ENATHUR, KANCHIPURAM _ 631 561

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

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Class : II B.E. (CSE)

Course Code: BCSF244AI2

Course Name: DATA SCIENCE TOOLKIT

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BONAFIDE CERTIFICATE

This is to certify that this is the bonafide record of work done by **Mr/Ms. S.L PAVAN KUMAR** _____,

with **Reg. No _11249A280** _____of **II-B.E.(CSE)** during the academic year 2025 – 2026.

Station:**ENATHUR**

Date:

Staff-in-charge **Head of the Department**

Submitted for the Practical examination held on_____.

Examiner-1 **Examiner-2**

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Ex No:1 **Date:05-02-26**

Introduction to Python Libraries for Data Science

Aim: To learn basic Python libraries for Data Science (NumPy, Pandas, Matplotlib, Scikit learn).

Program:

INPUT:

```
11249a435.py

import numpy as np

a = np.array([1,2,3,4,5])                                   #ch.viswa sri koushik
b = np.arange( start_or_stop: 1, stop: 10, step: 2)         #11249a435
c = np.linspace( start: 0, stop: 1, num: 5)
d = np.zeros((2,3))
e = np.ones((2,2))
f = np.eye(3)
g = np.random.randint( low: 1, high: 100,size=(3,3))

print("Array a:",a)
print("linspace c:",c)
print("zeros d:\n",d)
print("ones e:\n",e)
print("identity f:\n",f)
print("random g:\n",g)
```

OUTPUT:

```
C:\Users\Admin\PycharmProjects\PythonProject\.venv\Scripts\python.exe C:\Users\Admin\OneDrive\Desktop\PythonProject\11249a435.py
Array a: [1 2 3 4 5]
linspace c: [0.   0.25 0.5  0.75 1.  ]
zeros d:
 [[0. 0. 0.]
 [0. 0. 0.]]
ones e:
 [[1. 1.]
 [1. 1.]]
identity f:
 [[1. 0. 0.]
 [0. 1. 0.]
 [0. 0. 1.]]
random g:
 [[89 74  8]
 [47 52 35]
 [87 10 53]]

Process finished with exit code 0
```

INPUT:

11249a435.py

```python
import numpy as np
                                                                    # ch.viswa sri koushik
g = np.array([[78,97,29],                                            #11249a435
              [71,55,35],
              [2,72,98]])

print("shape:", g.shape)
print("dimensions:", g.ndim)
print("size:", g.size)
print("data type:", g.dtype
reshaped = g.reshape(9,1)
print("\nreshaped array:", reshaped)
print("Transpose:\n", g.T)

print("Flattened:", g.flatten())
```

OUTPUT:

```
C:\Users\Admin\PycharmProjects\PythonProject\.venv\Scripts\python.exe C:\Users\Admin\OneDrive\Desktop\PythonProject\11249a435.py
shape: (3, 3)
dimensions: 2
size: 9
data type: int64

reshaped array: [[78]
 [97]
 [29]
 [71]
 [55]
 [35]
 [ 2]
 [72]
 [98]]
Transpose:
 [[78 71  2]
 [97 55 72]
 [29 35 98]]
Flattened: [78 97 29 71 55 35  2 72 98]

Process finished with exit code 0
```

INPUT:

11249a435.py

```python
import numpy as np
                                                                    #CH.V SRI KOUSHIK VARDHAN
a = np.array([1,2,3,4,5])                                           #11249A435

print("Add:", np.add(a, x2: 2))
print("Square Root:", np.sqrt(a))
print("Power:", np.power(a, x2: 2))
print("Absolute:", np.abs([-1,-2,3]))

print("Mean:", np.mean(a))
print("Median:", np.median(a))
print("Standard Deviation:", np.std(a))
print("Variance:", np.var(a))
print("Minimum:", np.min(a))
print("Maximum:", np.max(a))
print("Sum:", np.sum(a))
print("Percentile (50):", np.percentile(a, q: 50))
```

OUTPUT:

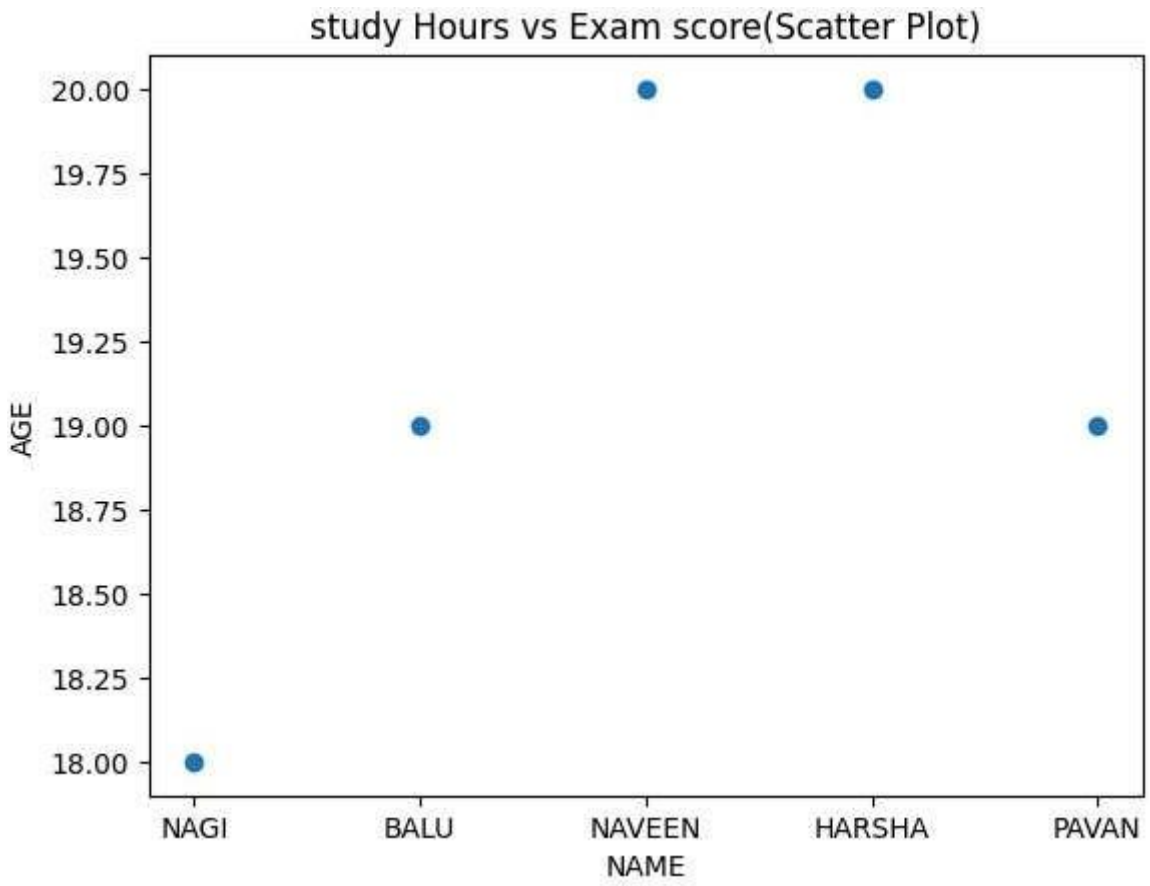
```
C:\Users\Admin\PycharmProjects\PythonProject\.venv\Scripts\python.exe C:\Users\Admin\OneDrive\Desktop\PythonProject\11249a435.py
Add: [3 4 5 6 7]
Square Root: [1.         1.41421356 1.73205081 2.         2.23606798]
Power: [ 1  4  9 16 25]
Absolute: [1 2 3]
Mean: 3.0
Median: 3.0
Standard Deviation: 1.4142135623730951
Variance: 2.0
Minimum: 1
Maximum: 5
Sum: 15
Percentile (50): 3.0

Process finished with exit code 0
```

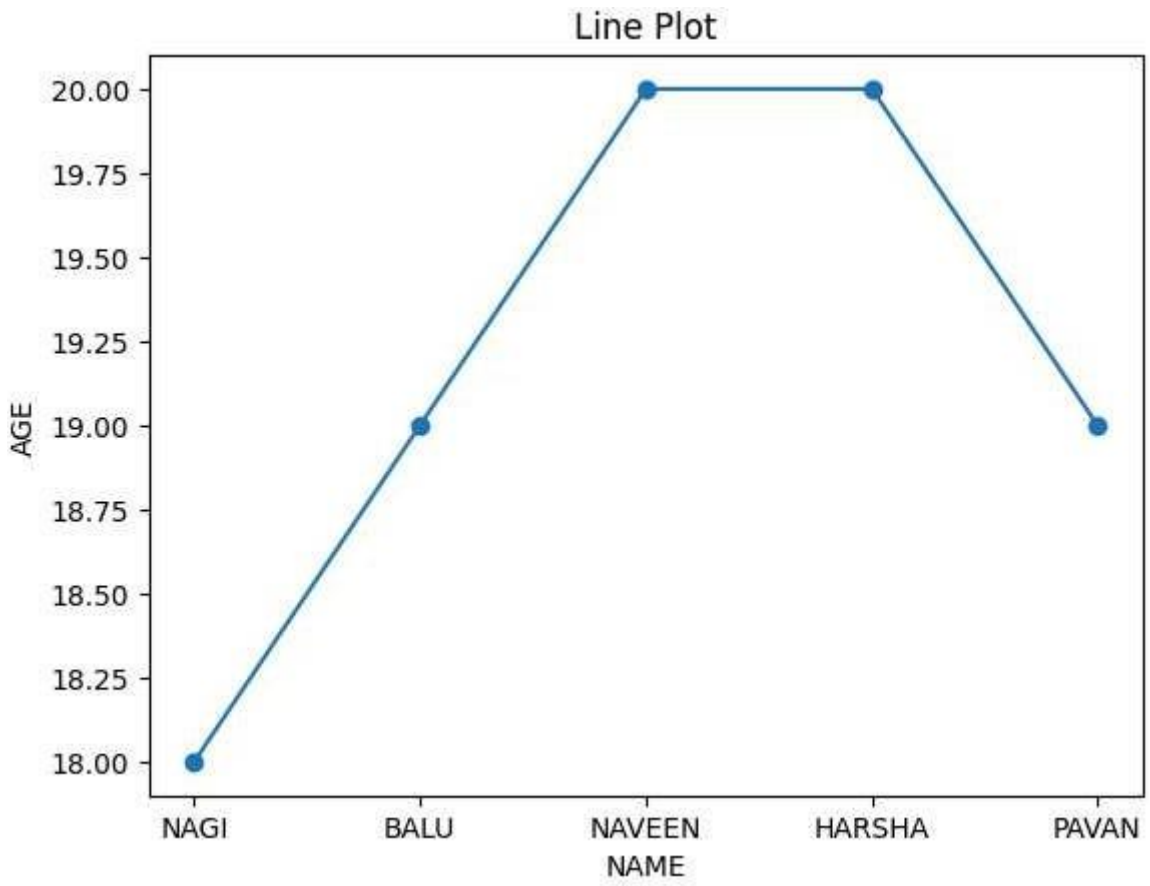
INPUT:

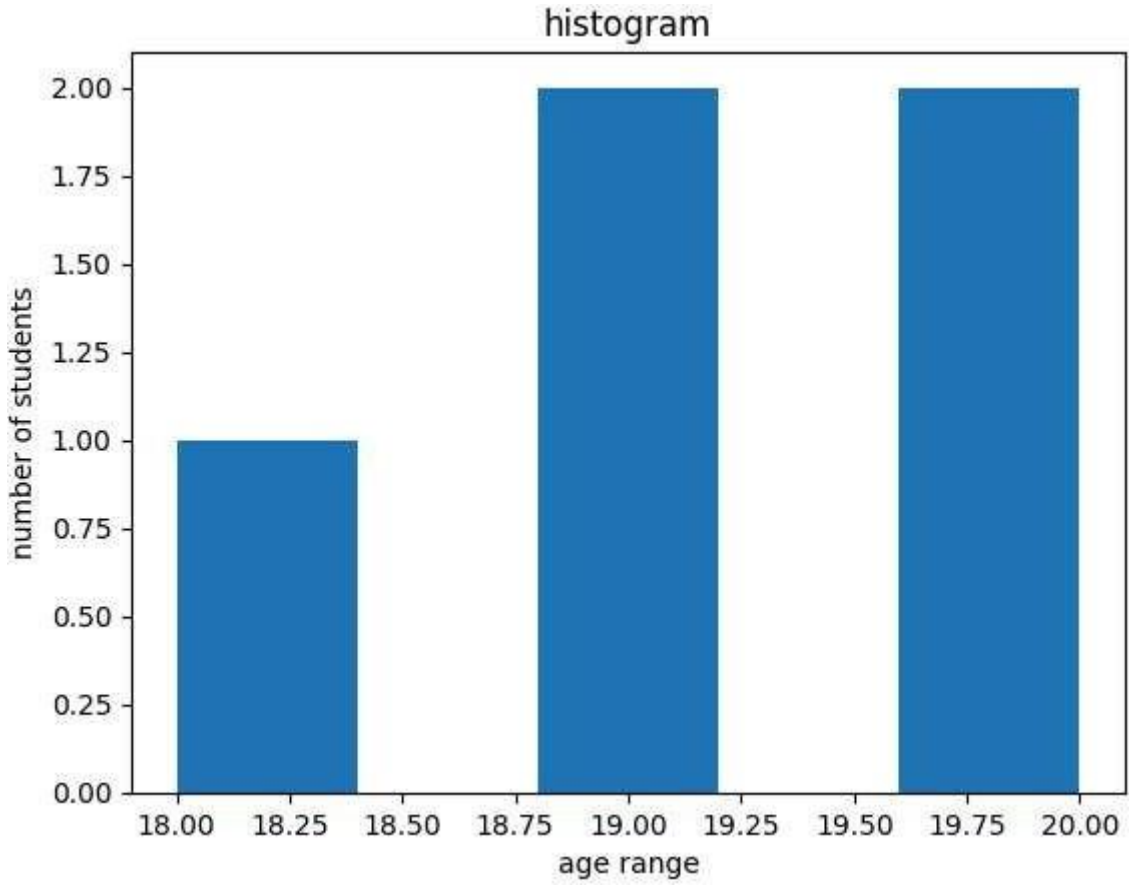
11249a435.py

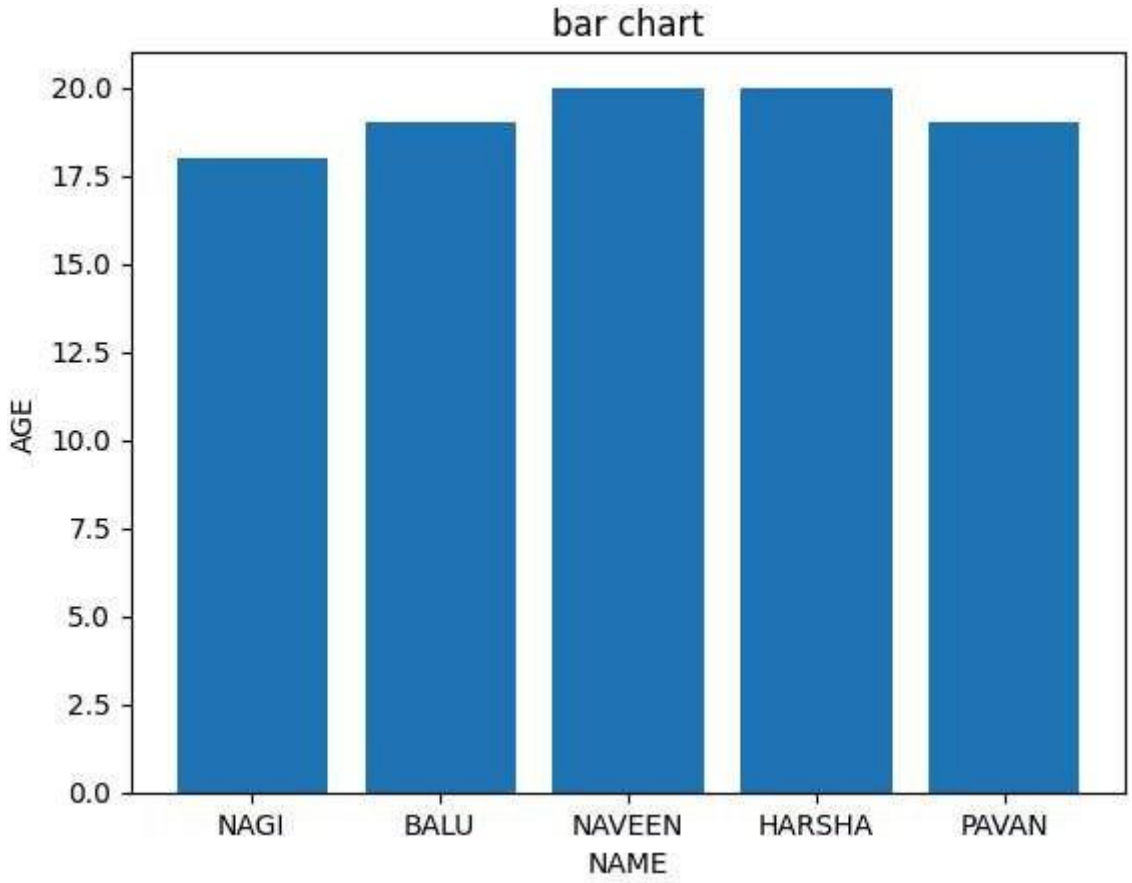
```python
import pandas as pd
import matplotlib.pyplot as plt                                    #CH.V SRI KOUSHIK VARDHAN
import numpy as np                                                 #11249A435

df = pd.read_csv('pv.csv')

plt.scatter(df['NAME'], df['AGE'])
plt.xlabel("NAME")
plt.ylabel("AGE")
plt.title("Scatter Plot")
plt.show()

plt.plot(df['NAME'], df['AGE'], marker='o')
plt.xlabel("NAME")
plt.ylabel("AGE")
plt.title("Line Plot")
plt.show()

plt.hist(df['AGE'], bins=5)
plt.xlabel("Age Range")
plt.ylabel("Number of Students")
plt.title("Histogram")
plt.show()

plt.bar(df['NAME'], df['AGE'])
plt.xlabel("NAME")
plt.ylabel("AGE")
plt.title("Bar Chart")
plt.show()
```

OUTPUT:


study Hours vs Exam score(Scatter Plot)
20.00
19.75
19.50
19.25
19.00
18.75
18.50
18.25
18.00
AGE
NAGI
BALU
NAVEEN
HARSHA
PAVAN
NAME


Line Plot
20.00
19.75
19.50
19.25
19.00
18.75
18.50
18.25
18.00
AGE
NAGI
BALU
NAVEEN
HARSHA
PAVAN
NAME


histogram
number of students
2.00
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00
18.00
18.25
18.50
18.75
19.00
19.25
19.50
19.75
20.00
age range


bar chart
20.0
17.5
15.0
12.5
10.0
7.5
5.0
2.5
0.0
AGE
NAGI
BALU
NAVEEN
HARSHA
PAVAN
NAME

INPUT:

11249a435.py

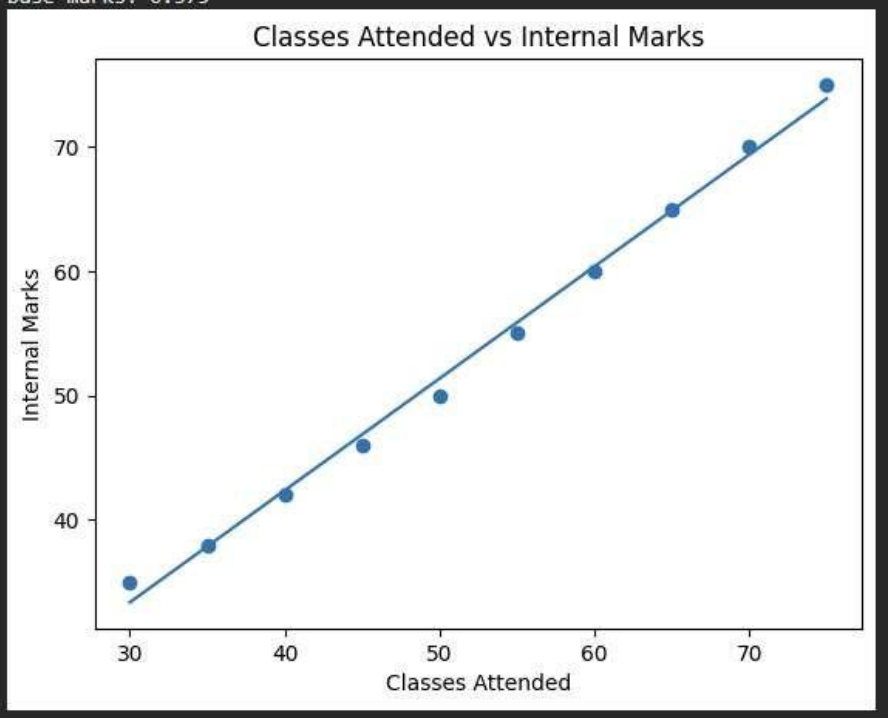
```python
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split                    #CH.V SRI KOUSHIK VARDHAN
from                                                  ssion             #11249A435
from                                                  or

df = pd.read_csv('/content/data.csv')

X = df[['Classes_Attended']]
y = df['Internal_Marks']

X_train, X_test, y_train, y_test = train_test_split( *arrays: X, y, test_size=0.2, random_state=7)

model = LinearRegression()
model.fit(X_train, y_train)

predictions = model.predict(X_test)

mse = mean_squared_error(y_test, predictions)

print("Mean Squared Error:", mse)
print("marks per class:", model.coef_[0])
print("Base marks:", model.intercept_)

plt.scatter(X, y)
plt.plot( *args: X, model.predict(X))
plt.xlabel("Classes Attended")
plt.ylabel("Internal Marks")
plt.title("Classes Attended vs Internal Marks")
plt.show()
```

Fetching Documentation...

Type hin

OUTPUT:

```
Mean Squared Error: 0.578125
marks per class: 0.9
Base marks: 6.375
```



Result: Basic operations using these libraries were performed successfully.

Ex No:02 **Date:12-02-26**

Perform Data exploration and preprocessing in Python

Aim: To perform data exploration and preprocessing in Python.

Program:

INPUT:

11249a435.py

```python
import pandas as pd
import numpy as np
from sklearn.preprocessing import StandardScaler
from sklearn.model_selection import train_test_split
                                                                    #CH.V SRI KOUSHIK VARDHAN
df = pd.read_csv('/content/data_output.csv')                        #11249A435
print("---Original raw Data---")
print(df)
print("\n")
print("---missing values count---")
print(df.isnull().sum())
print("\n")
df['Age'] = df['Age'].fillna(df['Age'].mode()[0])
df['Salary'] = df['Salary'].fillna(df['Salary'].mode()[0])
print("---DATA AFTER CLEANING---")
print(df)
print("\n")
df_encoded = pd.get_dummies(df, columns=['Country'])
df_encoded['Purchased'] = df_encoded['Purchased'].map({'Yes': 1, 'No': 0})
print("---DATA AFTER ENCODING---")
print(df_encoded)
print("\n")
X = df_encoded.drop( labels: 'Purchased', axis=1)
y = df_encoded['Purchased']
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
print("---FINAL PROCESSED DATA (Ready for AI)---")
print(pd.DataFrame(X_scaled, columns=X.columns).head())
```

Output:

```
   0   India  22.0  40000.0        No
   1     USA  25.0  60000.0       Yes
... 2   India   NaN  50000.0       Yes
   3     USA  30.0      NaN        No
   4      UK  28.0  72000.0       Yes
   5   India  35.0  58000.0       Yes


   ---missing values count---
   Country      0
   Age          1
   Salary       1
   Purchased    0
   dtype: int64


   ---DATA AFTER CLEANING---
     Country   Age   Salary Purchased
   0   India  22.0  40000.0        No
   1     USA  25.0  60000.0       Yes
   2   India  28.0  50000.0       Yes
   3     USA  30.0  60000.0        No
   4      UK  28.0  72000.0       Yes
   5   India  35.0  58000.0       Yes


   ---DATA AFTER ENCODING---
       Age   Salary  Purchased  Country_India  Country_UK  Country_USA
   0  22.0  40000.0          0           True       False        False
   1  25.0  60000.0          1          False       False         True
   2  28.0  50000.0          1           True       False        False
   3  30.0  60000.0          0          False       False         True
   4  28.0  72000.0          1          False        True        False
   5  35.0  58000.0          1           True       False        False


   ---FINAL PROCESSED DATA(Ready for ai)---
           Age    Salary  Country_India  Country_UK  Country_USA
   0 -1.484615 -1.693214            1.0   -0.447214    -0.707107
   1 -0.742307  0.338643           -1.0   -0.447214     1.414214
   2  0.000000 -0.677285            1.0   -0.447214    -0.707107
   3  0.494872  0.338643           -1.0   -0.447214     1.414214
   4  0.000000  1.557757           -1.0    2.236068    -0.707107
```

Result: Data was explored and preprocessed successfully using Python.

Ex No:03 **Date:19-02-26**

Implement regularised Linear regression

Aim: To implement regularized linear regression to improve model performance and control overfitting.

Program:

INPUT:

```python
from sklearn import datasets
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression, Ridge, Lasso                #CH.V SRI KOUSHIK VARDHAN
from sklearn.metrics import mean_squared_error                                 #11249A435

diabetes = datasets.load_diabetes()

X_train, X_test, y_train, y_test = train_test_split(
    *arrays: diabetes.data, diabetes.target, test_size=0.2, random_state=42
)

lr = LinearRegression().fit(X_train, y_train)
ridge = Ridge(alpha=1.0).fit(X_train, y_train)
lasso = Lasso(alpha=1.0).fit(X_train, y_train)

print(f"Linear Regression MSE: {mean_squared_error(y_test, lr.predict(X_test)):.2f}")
print(f"Ridge Regression MSE: {mean_squared_error(y_test, ridge.predict(X_test)):.2f}")
print(f"Lasso Regression MSE: {mean_squared_error(y_test, lasso.predict(X_test)):.2f}")
```

OUTPUT:

```
C:\Users\Admin\PycharmProjects\PythonProject\.venv\Scripts\python.exe C:\Users\Admin\OneDrive\Desktop\PythonProject\11249a435.py
Linear Regression MSE: 2900.19
Ridge Regression MSE: 3077.42
Lasso Regression MSE: 3403.58

Process finished with exit code 0
```

Result: The model was successfully implemented, and regularization helped reduce overfitting and improved prediction accuracy.

Ex No:04 **Date:05-02-26**

Implement Naive Bayes classifier for dataset stored as CSV file

Aim:

To implement the Naive Bayes classifier using a dataset stored in a CSV file for classification.

Program:

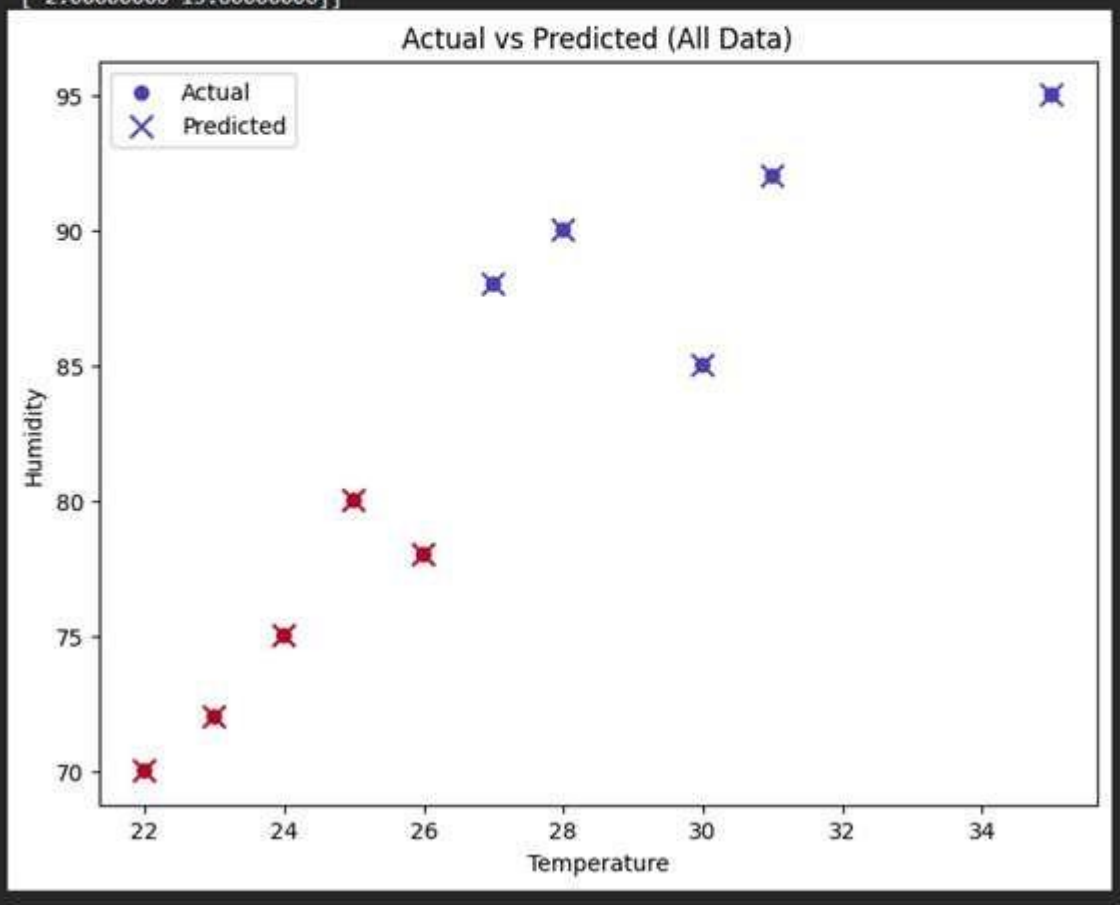
INPUT:

11249a435.py

```python
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import GaussianNB                                  #CH.V SRI KOUSHIK VARDHAN
from sklearn.preprocessing import LabelEncoder                              #11249a435
from sklearn.metrics import accuracy_score
import matplotlib.pyplot as plt
data = pd.read_csv('/content/temp_hum_play_data.csv')
le = LabelEncoder()
data['Play'] = le.fit_transform(data['Play'])
X = data[['Temperature', 'Humidity']]
y = data['Play']
X_train, X_test, y_train, y_test = train_test_split( *arrays: X, y, test_size=0.2, random_state=42)
model = GaussianNB()
model.fit(X_train, y_train)
prediction = model.predict(X_test)
all_prediction = model.predict(X)
print("Accuracy:", accuracy_score(y_test, prediction))
print("\nClass priors:\n", model.class_prior_)
print("\nMean (theta):\n", model.theta_)
print("\nVariance:\n", model.var_)
plt.figure(figsize=(8, 6))
plt.scatter(X['Temperature'], X['Humidity'], c=y,
            cmap='coolwarm',
            marker='o',
            label='Actual')
plt.scatter(X['Temperature'], X['Humidity'], c=all_prediction,
            cmap='coolwarm',
            marker='x',
            s=100,
            label='Predicted')
plt.xlabel("Temperature")
plt.ylabel("Humidity")
plt.title("Actual vs Predicted (All Data)")
plt.legend()
plt.show()
```

OUTPUT:

```
Accuracy: 1.0

class priors:
 [0.375 0.625]

Mean (theta) :
 [[30.66666667 89.33333333]
 [24.         75.        ]]

Variance:
 [[10.88888895 17.55555562]
 [ 2.00000006 13.60000006]]
```



Result:

The Naive Bayes model was successfully implemented, and it effectively classified the data with good accuracy.

Ex No:05 **Date:05-03-26**

Implement regularized logistic regression

Aim:

To implement regularized logistic regression for classification while reducing overfitting.

Program:

INPUT:

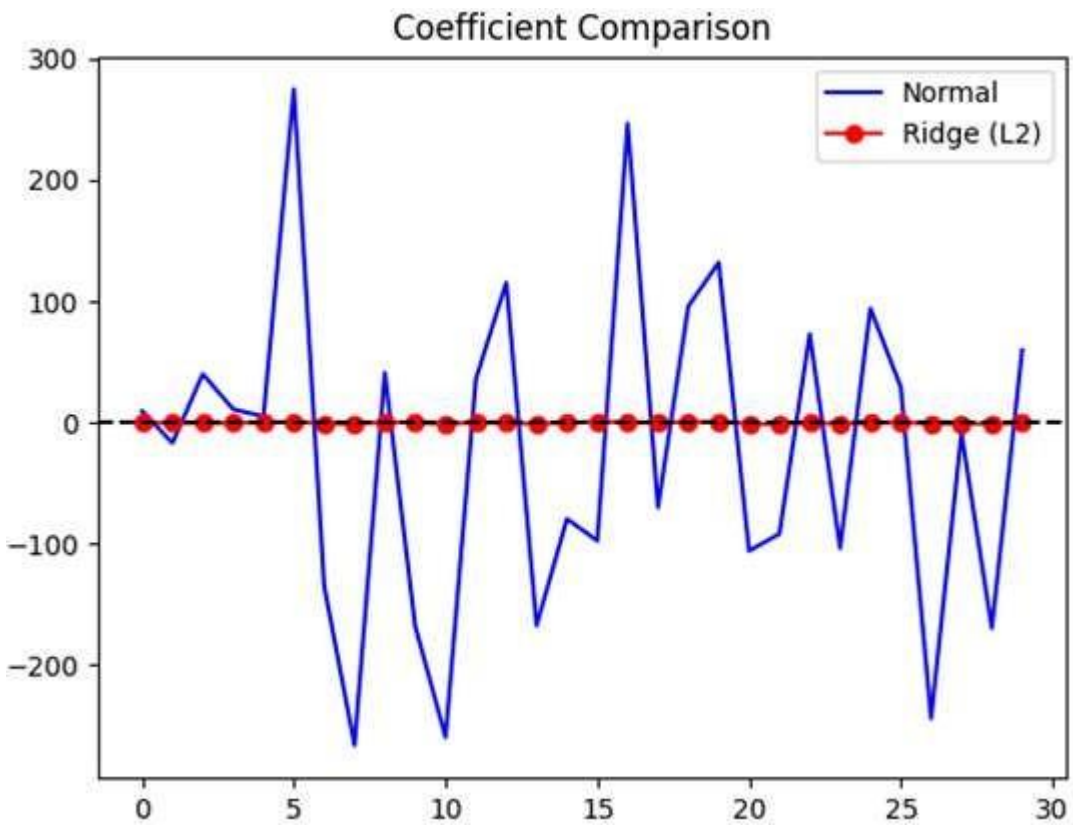
11249a435.py

```python
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.datasets import load_breast_cancer
from sklearn.linear_model import LogisticRegression
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import accuracy_score
data = load_breast_cancer()
X, y = data.data, data.target
features = data.feature_names
X_train, X_test, y_train, y_test = train_test_split(
    *arrays: X, y, test_size=0.2, random_state=42
)
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)
normal = LogisticRegression(penalty=None, max_iter=5000).fit(X_train, y_train)
normal_acc = accuracy_score(y_test, normal.predict(X_test))
ridge = LogisticRegression(penalty='l2', C=1.0, max_iter=5000).fit(X_train, y_train)
ridge_acc = accuracy_score(y_test, ridge.predict(X_test))
print(f"Normal Accuracy: {normal_acc:.3f}")
print(f"Ridge (L2) Accuracy: {ridge_acc:.3f}")
print("\nNormal Logistic Coefficients:")
print(pd.Series(np.round(normal.coef_[0], decimals=3), index=features))
print("\nRidge (L2) Coefficients:")
print(pd.Series(np.round(ridge.coef_[0], decimals=3), index=features))
plt.plot( *args: normal.coef_[0], label='Normal', color='blue')
plt.plot( *args: ridge.coef_[0], label='Ridge (L2)', color='red', marker='o')
plt.axhline(y=0, linestyle='--', color='black')
plt.legend()
plt.title('Coefficient Comparison')
plt.show()
```

Output:

```
Normal Accuracy: 0.939
Ridge (L2) Accuracy: 0.974

Normal Logistic Coefficients:
mean radius                   9.433
mean texture                 -16.986
mean perimeter                40.021
mean area                     10.890
mean smoothness                5.033
mean compactness             274.940
mean concavity              -134.838
mean concave points         -266.185
mean symmetry                 40.997
mean fractal dimension      -168.018
radius error                -259.639
texture error                 35.918
perimeter error              115.350
area error                  -167.389
smoothness error             -79.540
compactness error            -97.574
concavity error              246.608
concave points error         -70.052
symmetry error                95.954
fractal dimension error      131.939
worst radius                -105.871
worst texture                -91.921
worst perimeter               73.019
worst area                  -103.737
worst smoothness              93.975
worst compactness             29.285
worst concavity             -244.060
worst concave points          -8.908
worst symmetry              -169.655
```

```
worst fractal dimension     59.084
dtype: float64

Ridge (L2) Coefficients:
mean radius               -0.432
mean texture              -0.387
mean perimeter            -0.393
mean area                 -0.465
mean smoothness           -0.072
mean compactness           0.540
mean concavity            -0.801
mean concave points       -1.120
mean symmetry              0.236
mean fractal dimension     0.076
radius error              -1.268
texture error              0.189
perimeter error           -0.611
area error                -0.907
smoothness error          -0.313
compactness error          0.682
concavity error            0.175
concave points error      -0.311
symmetry error             0.500
fractal dimension error    0.616
worst radius              -0.880
worst texture             -1.351
worst perimeter           -0.589
worst area                -0.842
worst smoothness          -0.544
worst compactness          0.016
worst concavity           -0.943
worst concave points      -0.778
worst symmetry            -1.208
worst fractal dimension   -0.157
dtype: float64
```



Result:
The model was successfully implemented, and regularization improved accuracy and prevented overfitting.

Ex No:06 **Date:12-03-26**

Build models using different Ensembling techniques

Aim:

To build models using different ensembling techniques to improve prediction performance

Program:

INPUT:

11249a435.py

```python
from sklearn import datasets
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier, AdaBoostClassifier, StackingClassifier
from sklearn.svm import SVC
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score

iris = datasets.load_iris()

X_train, X_test, y_train, y_test = train_test_split(
    *arrays: iris.data, iris.target, test_size=0.2, random_state=42
)

rf = RandomForestClassifier(n_estimators=100).fit(X_train, y_train)
ada = AdaBoostClassifier(n_estimators=100).fit(X_train, y_train)

estimators = [
    ('rf', RandomForestClassifier()),
    ('svc', SVC(probability=True))
]

stack = StackingClassifier(
    estimators=estimators,
    final_estimator=LogisticRegression()
).fit(X_train, y_train)

print(f"Random Forest Accuracy: {accuracy_score(y_test, rf.predict(X_test)):.2f}")
print(f"AdaBoost Accuracy: {accuracy_score(y_test, ada.predict(X_test)):.2f}")
print(f"Stacking Accuracy: {accuracy_score(y_test, stack.predict(X_test)):.2f}")
```

Output:

```
C:\Users\Admin\PycharmProjects\PythonProject\.venv\Scripts\python.exe C:\Users\Admin\OneDrive\Desktop\PythonProject\11249a435.py
Random Forest Accuracy: 1.00
AdaBoost Accuracy: 0.93
Stacking Accuracy: 1.00

Process finished with exit code 0
```

Result:

The ensemble models were successfully implemented, and they provided better accuracy compared to individual models

Ex No:07 **Date:12-03-26**

Build models using Decision trees

Aim:

To build a model using Decision Trees for classification or prediction.

Program:

INPUT:

```python
from sklearn import datasets
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import accuracy_score
iris = datasets.load_iris()
X_train, X_test, y_train, y_test = train_test_split(
    *arrays: iris.data, iris.target, test_size=0.2, random_state=42
)
dt_clf = DecisionTreeClassifier(max_depth=3).fit(X_train, y_train)
print(f"Decision Tree Classifier Accuracy: {accuracy_score(y_test, dt_clf.predict(X_test)):.2f}")
print("Feature Importances:", dt_clf.feature_importances_)
```

Output:

```
C:\Users\Admin\PycharmProjects\PythonProject\.venv\Scripts\python.exe C:\Users\Admin\OneDrive\Desktop\PythonProject\11249a435.py
Decision Tree Classifier Accuracy: 1.00
Feature Importances: [0.         0.         0.39426873 0.60573127]

Process finished with exit code 0
```

Result:

The Decision Tree model was successfully implemented and provided good prediction performance.

Ex No:08 **Date:02-03-26**

Build model using SVM with different kernels

Aim:

To build an SVM model using different kernel functions for classification.

Program:

INPUT:

11249a435.py

```python
# Import libraries
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.svm import SVC
from sklearn.metrics import accuracy_score
from sklearn.datasets import load_iris
data = load_iris()
X = data.data
y = data.target
X_train, X_test, y_train, y_test = train_test_split(*arrays: X, y, test_size=0.2)
model_linear = SVC(kernel='linear')
model_linear.fit(X_train, y_train)
pred_linear = model_linear.predict(X_test)
acc_linear = accuracy_score(y_test, pred_linear)
print('Linear Kernel Accuracy:', acc_linear)
model_poly = SVC(kernel='poly', degree=3)
model_poly.fit(X_train, y_train)
pred_poly = model_poly.predict(X_test)
acc_poly = accuracy_score(y_test, pred_poly)
print('Polynomial Kernel Accuracy:', acc_poly)
model_rbf = SVC(kernel='rbf')
model_rbf.fit(X_train, y_train)
pred_rbf = model_rbf.predict(X_test)
acc_rbf = accuracy_score(y_test, pred_rbf)
print('RBF Kernel Accuracy:', acc_rbf)
```

Output:

```
Linear Kernel Accuracy: 1.0
Polynomial Kernel Accuracy: 0.9666666666666667
RBF Kernel Accuracy: 1.0

Process finished with exit code 0
```

Result:

The SVM models were successfully implemented with various kernels, and they achieved good classification performance.

Ex No:09 **Date:02-03-26**

Implement K-NN algorithm to classify a dataset

Aim: To implement the K-Nearest Neighbors (K-NN) algorithm to classify a dataset

Program:

INPUT:

11249a435.py

```python
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report
data = pd.read_csv("mobile_price.csv")
X = data.drop( labels: "price_range", axis=1)
y = data["price_range"]
X_train, X_test, y_train, y_test = train_test_split( *arrays: X, y, test_size=0.2, random_state=42)
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)
knn = KNeighborsClassifier(n_neighbors=3)
knn.fit(X_train, y_train)
y_pred = knn.predict(X_test)
print("Accuracy:", accuracy_score(y_test, y_pred))
print("Confusion Matrix:\n", confusion_matrix(y_test, y_pred))
print("Classification Report:\n", classification_report(y_test, y_pred))
```

Output:

```
Accuracy: 1.0
Confusion Matrix:
 [[1]]
Classification Report:
              precision    recall  f1-score   support

           2       1.00      1.00      1.00         1

    accuracy                           1.00         1
   macro avg       1.00      1.00      1.00         1
weighted avg       1.00      1.00      1.00         1
```

Result:

The K-NN model was successfully implemented and it classified the dataset with good accuracy

Ex No:10 **Date:02-03-26**

Build model to perform Clustering using K-means after applying PCA and determining

Aim:

To build a clustering model using K-Means after applying PCA for dimensionality reduction and determining optimal clusters.

Program:

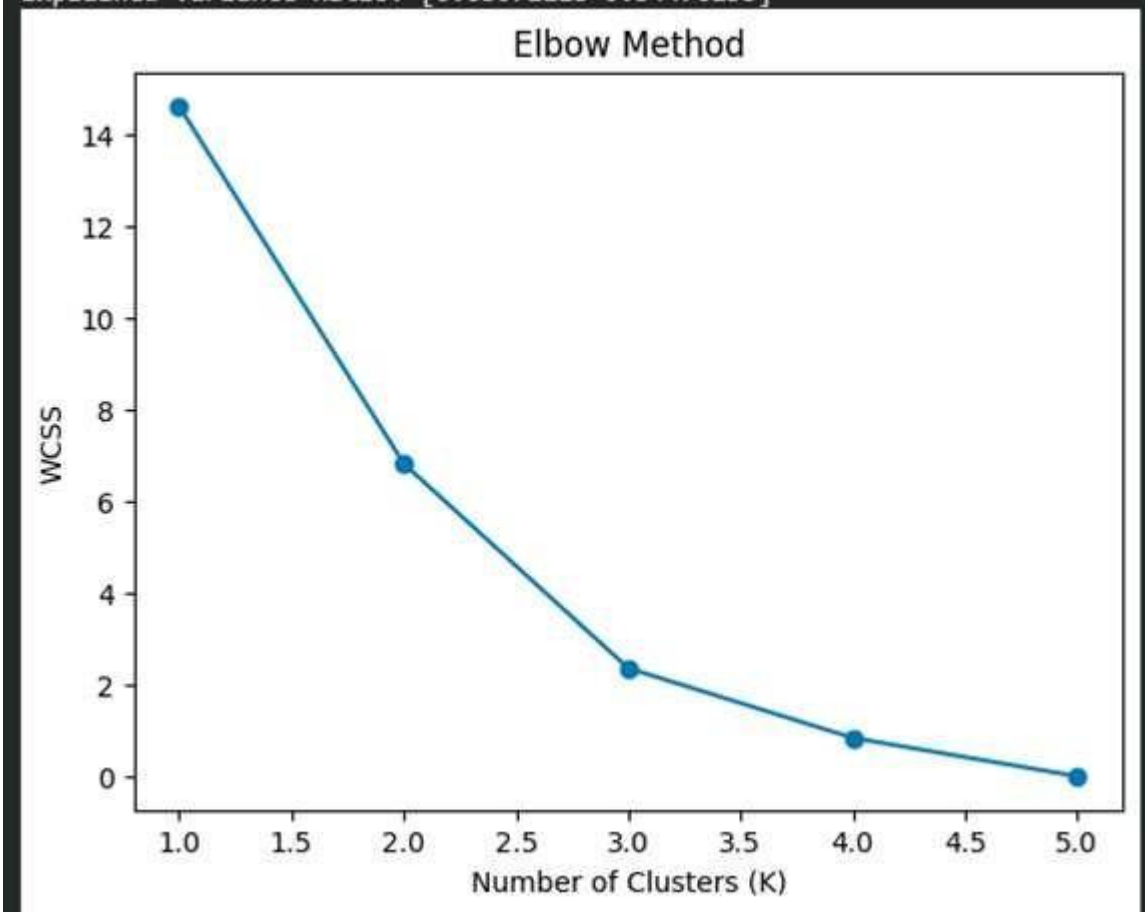
INPUT:

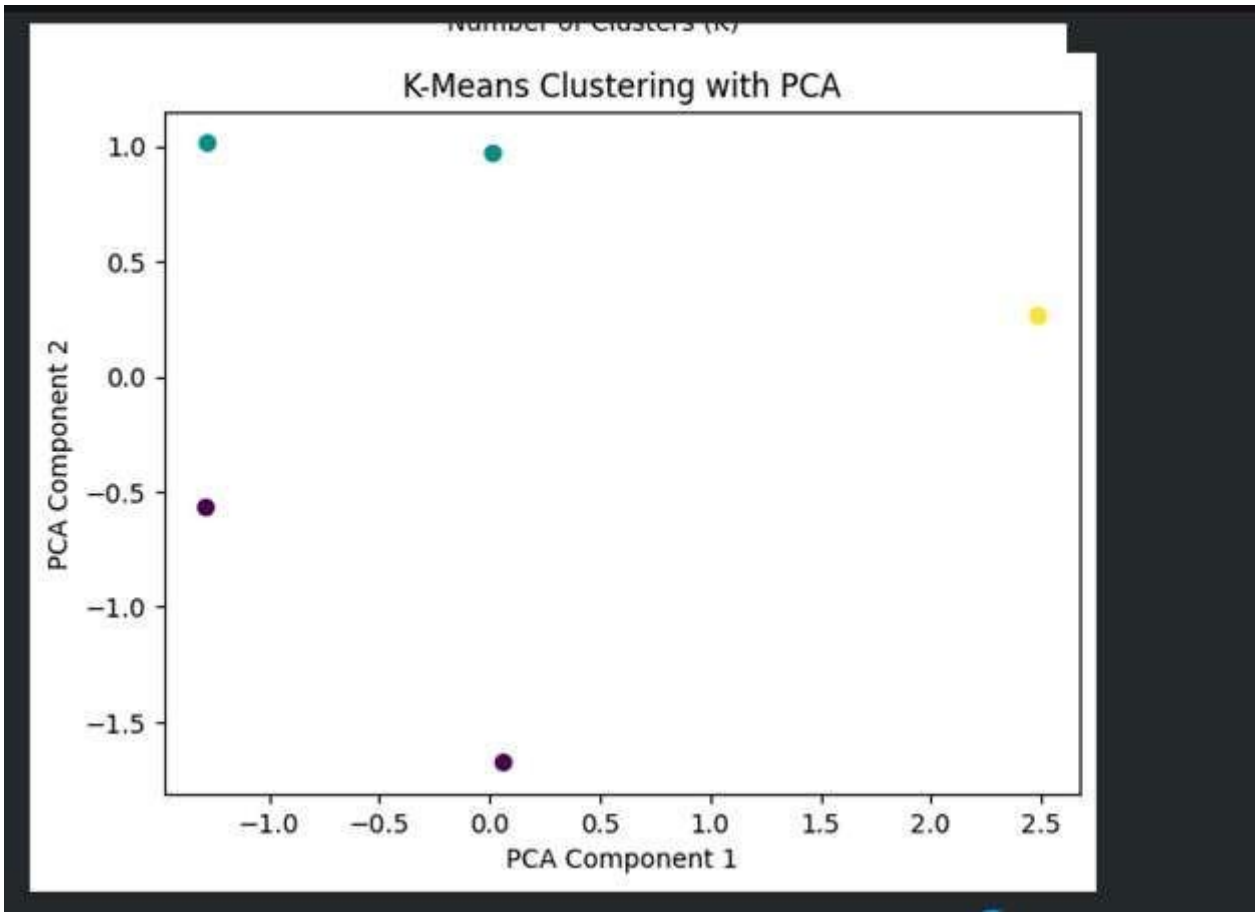
11249a435.py

```python
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA
from sklearn.cluster import KMeans
data = pd.read_csv("Mall_Customers.csv")
X = data[['Age', 'Annual Income (k$)', 'Spending Score (1-100)']]
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
pca = PCA(n_components=2)
X_pca = pca.fit_transform(X_scaled)
print("Explained Variance Ratio:", pca.explained_variance_ratio_)
wcss = []
max_k = min(10, len(X_pca))
for i in range(1, max_k + 1):
    kmeans = KMeans(n_clusters=i, random_state=0, n_init=10)
    kmeans.fit(X_pca)
    wcss.append(kmeans.inertia_)
plt.plot( *args: range(1, max_k + 1), wcss, marker='o')
plt.title("Elbow method")
plt.xlabel("Number of Clusters (k)")
plt.ylabel("WCSS")
plt.show()
optimal_k = min(3, len(X_pca))
kmeans = KMeans(n_clusters=optimal_k, random_state=0, n_init=10)
clusters = kmeans.fit_predict(X_pca)

plt.scatter(X_pca[:, 0], X_pca[:, 1], c=clusters, cmap='viridis')
plt.title('K-Means Clustering with PCA')
plt.xlabel('PCA Component 1')
plt.ylabel('PCA Component 2')
plt.show()
```

Output:

```
Explained Variance Ratio: [0.63071118 0.34476153]
```





Result:

The K-Means clustering model was successfully implemented after PCA, and the data was grouped effectively into clusters